Duality Gap in Dual Convex Optimization of ReLU Networks

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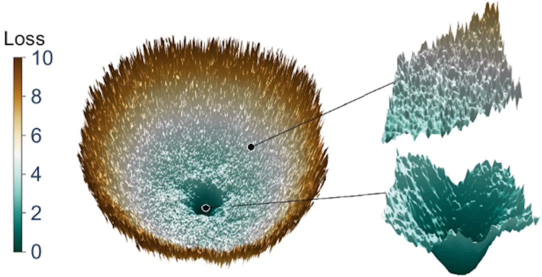
April 9, 2026

Presented at
SVC-SPARK Multidisciplinary International Conference 2026

Motivation: Why Does SGD Work?

- SGD was designed for convex problems
- We run it on provably non-convex losses
- It works anyway — why?
- Common explanation: overparameterization and favourable initialisation
- **A more precise answer emerged in 2020**



Duality Gap: Key Definitions

Every optimization problem has a companion **dual**.

$$P^* \geq D^*$$

primal (hard problem) — dual (companion)

$$\text{gap} = P^* - D^*$$

$$\textbf{strong duality} \iff \text{gap} = 0$$

For convex problems the gap is zero. For non-convex — usually not.

For ReLU networks with weight decay, the gap turns out to be zero.

Convex Reformulation: Pilanci & Ergen (2020)

Non-convex primal:

$$\min_{\{\mathbf{u}_j, \alpha_j\}_{j=1}^m} \frac{1}{2} \left\| \sum_{j=1}^{m} (\mathbf{X}\mathbf{u}_j)_+ \alpha_j - \mathbf{y} \right\|_2^2 + \frac{\beta}{2} \sum_{j=1}^{m} \left(\|\mathbf{u}_j\|_2^2 + \alpha_j^2 \right) \qquad \text{(primal)}$$

$\Downarrow$ **zero duality gap**

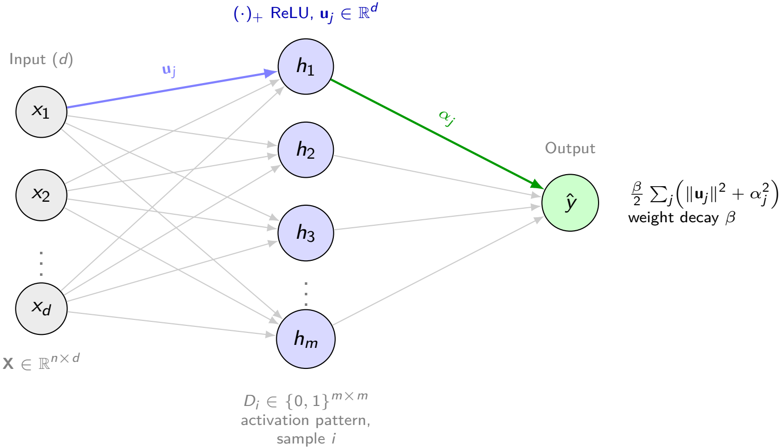
Equivalent convex program (SOCP):

- $O((n/r)^r)$ variables; finite, exact
- Width condition: $m \geq m^* \leq n + 1$

Theorem 1 [Pilanci & Ergen, 2020]

Two-layer ReLU + weight decay admits an exact convex reformulation. Strong duality holds for $m \geq m^*$.

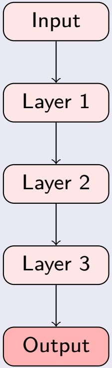
Architecture of the Primal Two-Layer ReLU Network



Two-layer ReLU network: m hidden neurons, n training samples, $P = O((n/r)^r)$ distinct activation patterns D_i.

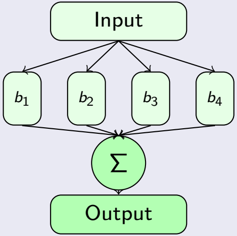
Depth vs. Architecture: Effect on the Duality Gap

Standard L-layer network



$P^* > D^*$ for $L \geq 3$

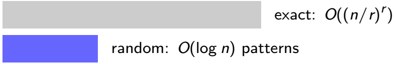
Parallel (K branches) network



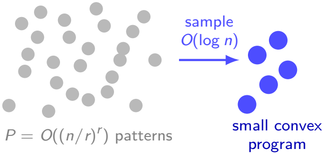
$P^* = D^*$ for **any** L

Architecture, not depth, determines the duality gap.

Complexity comparison:



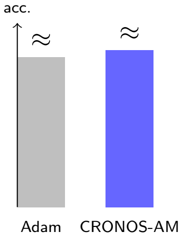
Random sampling idea:



Approximation guarantee (Thm. 2.1): $P^* \leq \tilde{p}^* \leq C\sqrt{\log 2n}\, P^*$

First poly-time approximation guarantee for regularized ReLU networks.

Adam vs. CRONOS-AM on ImageNet:



JAX
RTX-4090
ADMM
ImageNet

- Matches/beats tuned Adam
- Essentially no hyperparameters
- Global opt. guarantee (2-layer)

ImageNet: 1000 classes, 1.2M training images

Takeaway

ReLU network training was never really non-convex —

and whether the hidden convex program is exact, tractable, and practically solvable depends on three things:

width | architecture | log n

width	**architecture**	**log n**
Pilanci-Ergen 2020	Wang-Ergen-Pilanci 2021	Kim-Pilanci 2024
$m \geq m^*$	parallel branches	$\sqrt{\log n}$ approximation

Thank you — questions welcome.

References I

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Dwaraknath, Rajat Vadiraj, Tolga Ergen, and Mert Pilanci (2023). "Fixing the NTK: From Neural Network Linearizations to Exact Convex Programs". In: URL: https://arxiv.org/abs/2309.15096.

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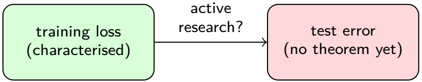
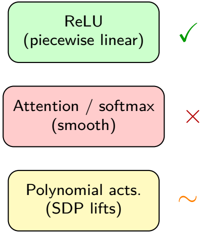
B1: Generalisation, Not Training Loss

- Zero duality gap = global optimum of **training loss**
- Test error / generalisation bounds: **NOT** covered
- Optimal-set polytope → hope for future generalisation bounds
- The most important honest thing to say about this work



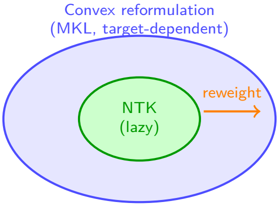
B2: Does This Extend to Transformers?

- Convex reformulation requires ReLU: piecewise linear
- Attention (softmax) is smooth — no hyperplane patterns
- Extension to transformers: **explicitly open** (, 2024 conclusion)
- Polynomial activations: partial via SDP (, 2021)



B3: Relationship to NTK

- NTK (Jacot 2018): infinite width, no feature learning
- Convex reformulation: exact at finite width, with feature learning
- (2023):
  NTK is a special case of the MKL structure
- Iterative reweighting upgrades NTK to the full convex solution
- feature learning = the gap between NTK and full convex solution



feature learning = the gap

B4: What About ResNets?

- ResNet residual block = 2-branch parallel network
- Branch 1: **identity** path
- Branch 2: **transform** path (Conv $\to$ BN $\to$ ReLU)
- (2021): ResNets + weight decay $\subseteq$ parallel-network framework
- Strong duality restored: inherits zero-duality-gap result